

Peijie (Max) Ma

✉ pm3688@nyu.edu 📞 +1 (516) 288-0844 in [LinkedIn](#) 🌟 [Tau Beta Pi \(Honor Society\)](#) 📄 [Google Scholar \(278+ citations\)](#)

Education

New York University

Sept. 2023 – June 2027 (Expected)

- **B.S. Double Major in Computer Engineering and Mathematics GPA: 3.977/4.0**
- **Coursework:** Optimization, Complex Variables, Embedded Systems, Object-Oriented Programming, Signals and Systems, Digital Logic, Computer Architecture, Electronics, Probability, Statistics, Machine Learning, Data Structures and Algorithms, Linear Algebra
- **Honors & Awards:** Dean's List (2024–2026), The Myron M. Rosenthal Merit Award (2026, awarded to juniors with the highest academic standing)

Research Interests

My research focuses on machine learning systems for physical-world modeling, with particular interests in 3D scene understanding, geometry-aware algorithms, and efficient, hardware-aware deployment.

Research Experience

Furniture-Aware Floorplan Recognition & One-Click 3D Reconstruction

Jan. 2026 - Present

Supervisor: [Prof. Nikola Janjušević](#) [🔗](#)

- Developed an end-to-end furniture-aware 2D-to-3D reconstruction pipeline that transforms raster floorplans into semantically populated, simulation-ready 3D digital twins.
- Integrated multi-task CNN floorplan parsing with an enclosure-preserving wall repair algorithm and geometric vectorization using Zhang–Suen thinning, RDP simplification, Harris corner detection, and PCA-based orientation snapping.
- Built a Python/Blender synthesis engine that calibrates metric scale, generates watertight structural meshes, carves door and window openings, and instantiates semantic furniture assets from structured JSON representations.

NYURay: Python-Based Ray Tracer for Wireless Channel Prediction

Sept. 2024 – Present

Research Advisor: [Prof. Theodore S. Rappaport](#) [🔗](#) (Founder of NYU WIRELESS)

- Developed site-specific ray-tracing validation workflows by aligning simulated propagation paths with measured power delay profiles at upper mid-band frequencies.
- Implemented a multi-stage transmitter and receiver location optimization method based on power delay profile alignment to reduce geometric registration errors in ray-tracing simulations.
- Built automated Python and Blender workflows for TX-RX placement, 3D scene processing, and large-scale comparison between NYURay predictions and real-world wireless measurements.

Comprehensive FR3 Radio Propagation Measurement and Modeling

Nov. 2023 - Present

Supervisor: [Prof. Theodore S. Rappaport](#) [🔗](#)

- Contributed to the world's first comprehensive upper mid-band measurement campaign at 6.75 GHz and 16.95 GHz for 5G/6G wireless systems.
- Developed a detailed statistical channel model across over 100 TX-RX location pairs.
- Processed and analyzed 150+ GB of power delay profiles to enable large-scale data-driven modeling.

AMAR: Automated Material-Aware 3D Reconstruction

Sept. 2023 - Present

Supervisors: [Prof. David Fouhey](#) [🔗](#) & [Prof. Theodore S. Rappaport](#) [🔗](#)

- Developed a machine-learning-assisted material-aware 3D reconstruction framework for wireless ray tracing, achieving detection accuracy within 10 cm and classification precision exceeding 89%.
- Validated and calibrated the NYURay ray tracer using 30 different objects with 10 material types for accurate ray-tracing simulations.
- Reconstructed 3D digital twins of an indoor environment (2000 m²) and an outdoor environment (2 km²) using Blender, enabling accurate hardware-aware wireless system modeling and simulation validation.

Teaching Experience

Teaching Assistant *Intro to ECE 1002*

Feb. 2025 - Present

- Assisted in authoring a comprehensive textbook about the historical evolution, fundamental principles, and modern advancements of Electrical & Computer Engineering and Computer Science.
- Designed and published 13 weekly assignments aligned with lecture content together with detailed solutions.
- Provided academic mentorship to 160 students through interactive and personalized support.

Professional Experience

Test Engineer *Mini-Circuits*

May. 2025 - Sept. 2025

- Analyzed RF circuit schematics, characterized microwave components for proper functionality and stress derating using professional RF measurement equipment such as ANSYS, Draftsight, and ADS.
- Supported E-Run final design verification and H-Run specification development over 20 commercial products based on customer requirements.
- Used spectrum-analyzer to measure S-parameters, noise-figure, Psat, P1dB, and stability (add something about the reason why you do this or the outcomes).

Algorithm Engineer *Matrixtime Robotics Co. Ltd, Shanghai, China*

Dec. 2023 – Feb. 2024

- Standardized functions for a semiconductor defect inspection program, increasing accuracy to 92%.
- Optimized data acquisition algorithms, reducing processing time by 42% and improving chip yield detection efficiency by 85.1%.

Assistant Data Engineer

Shanghai, China

Shanghai Det Vista Co. Ltd

Jun 2023 - Sept 2023

- Authored, tested, and enhanced over 2,000 lines of code for software applications, improving system performance and user interface ratings by 40%.
- Managed the development and maintenance of databases serving 10,000+ users, supporting key community programs across China.
- Spearheaded the launch of a community-driven mobile application, attracting 3,400+ downloads within the first two months.

Selected Publications

- [1] M. Ying, G. Qian, X. Wang, **P. Ma**, D. Shakya, T.S. Rappaport, "HoRAMA: Holistic Reconstruction with Automated Material Assignment for Ray Tracing using NYURay," in *IEEE International Conference on Communications (ICC)*, TGlaskow, UK, Jun. 2026, pp. 1–6, [PDF] [↗](#)
- [2] M. Ying, **P. Ma**, D. Shakya, T.S. Rappaport, "Multi-Stage Location Optimization Through Power Delay Profile Alignment Using Site-Specific Wireless Ray Tracing," in *GLOBECOM 2025 - 2025 IEEE Global Communications Conference*, Taipei, Taiwan, 2025, pp. 5169-5174, doi: [10.1109/GLOBECOM59602.2025.11432293](https://doi.org/10.1109/GLOBECOM59602.2025.11432293) [↗](#)
- [3] M. Ying, D. Shakya, T. S. Rappaport, **P. Ma**, et al., "Comprehensive FR1(C) and FR3 Lower and Upper Mid-Band Propagation and Material Penetration Loss Measurements and Channel Models in Indoor Environment for 5G and 6G," in *IEEE Open Journal of the Communications Society*, vol. 5, pp. 5192-5218, 2024, doi: [10.1109/OJCOMS.2024.3431686](https://doi.org/10.1109/OJCOMS.2024.3431686) [↗](#)
- [4] M. Ying, D. Shakya, **P. Ma**, G. Qian, and T. S. Rappaport, "Site-Specific Location Calibration and Validation of Ray-Tracing Simulator NYURay at Upper Mid-Band Frequencies," *npj Wireless Technology*, vol. 2, no. 1, Art. no. 8, 2026.[PDF] [↗](#)

Technical Skills

Programming: Python, C/C++, Java, MATLAB, SQL, Verilog, LaTeX

Machine Learning & Computer Vision: PyTorch, OpenCV, CNNs, Semantic Segmentation, Object Detection

3D Modeling & Scientific Computing: Blender, Open3D, NumPy, SciPy

Engineering Tools: ANSYS, ADS, AutoCAD, Fusion 360, Spectrum Analyzer

Research Areas: Machine Learning Systems, 3D Scene Understanding, Digital Twins, Ray Tracing, Wireless Channel Modeling